Black-Scholes Derivation via Expectation

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We consider the forward process described by the SDE

$$\frac{dS_t}{S_t} = r\,dt + \sigma\,dW_t$$

With solution through Itô's lemma, using $\tau = T - t$

$$S_T = S_t \exp((r - \frac{\sigma^2}{2})\tau + \sigma W_T)$$

As $W_T \sim N(0, T)$, we can define $W_T = \sqrt{T}\,Z_T$, where $Z_T \sim N(0, 1)$, and thus

$$S_T = S_t \exp((r - \frac{\sigma^2}{2})\,\tau + \sigma\sqrt{\tau}\,Z_T) \tag{1}$$

Under this setup, we then consider the expectation under the $\mathbb{Q}$-measure to assess the value of a European call option

$$C(\cdot) = e^{-r\tau}\,\mathbb{E}[(S_T - K)^+]$$

Further,

$$\mathbb{E}[(S_T - K)^+] = \mathbb{E}[S_T - K \cdot \mathbb{I}_{S_T > K}]$$

$$= \mathbb{E}[S_T \cdot \mathbb{I}_{S_T > K}] - \mathbb{E}[K \cdot \mathbb{I}_{S_T > K}] \tag{2}$$

Concerning the second term,

$$\mathbb{E}[K \cdot \mathbb{I}_{S_T > K}]$$

We consider,

$$S_T > K \Longrightarrow \ln S_T > \ln K$$

And using (1),

$$\ln S_t \ + (r - \frac{\sigma^2}{2})\, \tau \ + \sigma \, \sqrt{\tau} \, Z_T > \ln K$$

$$Z_T > \frac{\ln{(K/S_t)} - (r - \frac{\sigma^2}{2})\, \tau}{\sigma \, \sqrt{\tau}} = d_2$$

And hence,

$$\mathbb{I}_{S_T > K} = \mathbb{I}_{Z_T > d_2} \tag{3}$$

As such,

$$\mathbb{E}[K \cdot \mathbb{I}_{S_T > K}] = \mathbb{E}[K \cdot \mathbb{I}_{Z_T > d_2}]$$

$$\int_0^\infty K \cdot \mathbb{I}_{S_T > K} \, f(s) \, ds = \int_0^\infty K \cdot \mathbb{I}_{Z_T > d_2} \, g(z) \, dz$$

$$\int_K^\infty K \, f(s) \, ds = \int_{d_2}^\infty K \, g(z) \, dz$$

While we do not know $f(s)$, we know that $g(z) = \phi(z)$ is the PDF of the standard normal, by definition of Z. And hence,

$$\mathbb{E}[K \cdot \mathbb{I}_{S_T > K}] = K \int_{d_2}^\infty \phi(z) \, dz = K \, \Phi(-d_2) \tag{4}$$

Concerning the first term of (2),

$$\mathbb{E}[S_T \cdot \mathbb{I}_{S_T > K}]$$

Using (1) and (3),

$$\mathbb{E}[S_T \cdot \mathbb{I}_{S_T > K}] = \mathbb{E}[S_t \exp((r - \frac{\sigma^2}{2})\, \tau + \sigma\sqrt{\tau}\; Z_T) \cdot \mathbb{I}_{Z_T > d_2}]$$

Grouping deterministic elements,

$$\mathbb{E}[S_T \cdot \mathbb{I}_{S_T > K}] = S_t \; e^{(r - \frac{\sigma^2}{2})\tau} \; \mathbb{E}[e^{\sigma\sqrt{\tau} Z_T} \cdot \mathbb{I}_{Z_T > d_2}]$$

$$= S_t \; e^{(r - \frac{\sigma^2}{2})\tau} \int_{d_2}^{\infty} e^{\sigma\sqrt{\tau} z} \phi(z)\, dz = S_t \; e^{(r - \frac{\sigma^2}{2})\tau} \int_{d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\sigma\sqrt{\tau} z - \frac{z^2}{2}} \; dz$$

Critically, $\frac{1}{\sqrt{2\pi}} e^{\sigma\sqrt{\tau} z - \frac{z^2}{2}}$ can be written as $\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - \sigma\sqrt{\tau})^2} \cdot e^{\frac{\sigma^2 \tau}{2}}$, which describes the PDF of a normal distribution shifted by $\sigma\sqrt{\tau}$, scaled by the constant $e^{\sigma^2 \tau / 2}$. As such,

$$\mathbb{E}[S_T \cdot \mathbb{I}_{S_T > K}] = S_t \; e^{(r - \frac{\sigma^2}{2})\tau} e^{\frac{\sigma^2 \tau}{2}} \int_{d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - \sigma\sqrt{\tau})^2} \; dz$$

$$= S_t \; e^{r\tau} \Phi(-(d_2 - \sigma\sqrt{\tau}))$$

$$\mathbb{E}[S_T \cdot \mathbb{I}_{S_T > K}] = S_t \; e^{r\tau} \Phi(d_1), \tag{5}$$

With $d_1 = -d_2 \; + \sigma\sqrt{\tau}$.

Lastly, using (2), (4), and (5),

$$\mathbb{E}[(S_T - K)^+] = S_t \; e^{r\tau} \Phi(d_1) - K \; \Phi(-d_2)$$

$$C(\cdot) = e^{-r\tau} \; \mathbb{E}[(S_T - K)^+]$$

$$C(\cdot) = S_t \; \Phi(d_1) - K e^{-r\tau} \; \Phi(-d_2) \tag{6}$$